F1 Top-10 Finish Prediction Machine Learning Classification Report

CX016-2.5-3-IML - Introduction to Machine Learning

Group members	TP number
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Module code	CX016-2.5-3-IML
Module title	Introduction to Machine Learning
Class / intake code	CSSE___CX016-2.5-3-IML-L-1___2026-01-30
Hand out date	13 February 2026
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Weightage	30%
Deliverable	Report, notebook, dataset, scripts and submission ZIP
Main task	Predict whether each Formula 1 driver finishes in the top 10
Dataset	6,999 driver-race rows, 201 variables, seasons 2010-2026
Primary model	Random Forest classifier

Temporal validation: 2025 holdout | *Dataset: 2010-2026* | *Target: top10_finish*

Submission Details

Field	Detail
Group member	Cavaignac Romain - TP1458
Group member	Dubernet Mathieu - TP145868
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Table of Contents

Abstract

This project investigates whether machine learning can predict if a Formula 1 driver will finish a Grand Prix in the top 10. This is a useful sporting analytics problem because top-10 classification is directly linked to points scoring, race strategy and team performance evaluation. A single model-ready dataset was built from public Formula 1 data covering 2010 to 2026, combining race results, qualifying, standings, circuit information, historical pit-stop indicators, weather, and selected FastF1-derived features. The dataset was cleaned, merged and transformed into 6,999 driver-race observations with 201 variables and no missing values. The main predictive task is binary classification with `top10_finish` as the target. Logistic regression, random forest, extra trees, histogram gradient boosting and a multilayer perceptron were compared using temporal validation. The best holdout model is random forest, with 2025 race precision@10 of 0.775. A separate finish-position regressor was also trained as a ranking extension for predicted finishing order, but the primary assignment result remains the comparison and validation of predictive models on the tabular top-10 classification task.

Introduction, Aim and Objectives

Formula 1 results are influenced by many interacting factors: car performance, driver ability, qualifying position, reliability, circuit type, weather and race strategy. This makes the problem more challenging than a simple ranking by previous points. The goal of this project is to use historical and pre-race information to estimate whether each driver will finish inside the top 10.

The aim is to build, evaluate and compare machine learning models for Formula 1 top-10 finish prediction.

The objectives are:

- collect and consolidate a reasonably large, non-trivial Formula 1 dataset
- include both numeric and categorical features
- perform exploratory data analysis and preprocessing
- engineer predictive features that avoid direct post-race leakage
- train and compare at least two predictive machine learning models
- include model tuning and validation
- interpret the results and recommend the most suitable model
- provide reproducible code, generated figures and a final submission archive

The scope is supervised prediction at driver-race level. The main output is a probability and ranking of likely top-10 finishers. Exact race simulation, live telemetry modelling and lap-by-lap race strategy optimization are treated as future work rather than core assignment requirements.

Assignment Compliance Snapshot

The assignment question requires one challenging dataset, predictive models, preprocessing, EDA, tuning, validation, interpretation, a report and source code. The current project satisfies these requirements as follows.

Assignment requirement	Project evidence	Status
One dataset	`data/final/f1_top10_model_dataset.csv` is the single model-ready dataset used for modelling	Satisfied
Reasonable size	6,999 driver-race rows, 201 variables and seasons 2010-2026	Satisfied
Not perfectly clean	Raw API tables require identifier matching, missing-data handling, external weather joins and partial FastF1 coverage	Satisfied
Numeric and categorical variables	Final dataset includes 179 numeric/bool columns and 22 categorical/text columns	Satisfied
More than 12 variables	The final dataset has 201 columns	Satisfied
At least two predictive models	Logistic regression, random forest, extra trees, histogram gradient boosting and MLP were compared	Exceeded
Dataset preparation and EDA	Figures in `outputs/figures/` discuss target balance, season coverage, grid/finish relation and missing values	Satisfied
Optimization/tuning	MLP tuning, fixed ensemble hyperparameters and temporal validation are documented	Satisfied

Assignment requirement	Project evidence	Status
Model validation	2025 holdout plus expanding-window rolling backtest	Satisfied
Report and compressed submission	`report/Report.docx`, `report/Report.pdf` and `submission/IML_Assignment_GroupX.zip`	Satisfied
Source code in notebook format	`notebooks/ML_Project_Code.ipynb` provides the assignment-facing notebook	Satisfied

Related Works

Previous Formula 1 analytics work often focuses on lap-time prediction, race outcome prediction, tire degradation or separating driver and constructor effects. These studies informed the feature choices in this project: qualifying position and team strength are important baseline predictors, while circuit, weather, tire and historical form features can improve the model.

Source	Focus	Relevance to this project
Breiman (2001)	Random forests	Motivates using tree ensembles for mixed tabular data
Friedman (2001)	Gradient boosting machines	Supports boosting as a strong tabular baseline
Pedregosa et al. (2011)	scikit-learn	Main software library used for modelling pipelines
Nigro (2020)	F1 race predictor	Shows practical race prediction using historical F1 data
Zhao (2024)	Deep neural network lap-time forecasting	Shows neural networks can model F1 performance signals
Jafri (2024)	F1 race outcome prediction	Uses historical race data for race prediction
Noe and Patel (2024)	Telemetry-based lap prediction	Motivates lap and performance features

Source	Focus	Relevance to this project
Cappello and Hoegh (2026)	Tire degradation modelling	Supports the value of tire and strategy variables
Jolpica F1 API	Ergast-compatible data source	Provides race, qualifying, standings and pit-stop data
FastF1	Timing, lap and race-control data	Optional enrichment for recent seasons
Open-Meteo Archive	Historical weather	Adds race-day weather context

Compared with much of the related work, this project does not try to predict the race winner only. It predicts top-10 classification for every driver-race row, which creates a larger supervised dataset and aligns with the points-scoring structure of modern F1. Because the public studies use different seasons, targets and feature access, their exact scores are not directly comparable; this report therefore compares against their modelling choices and validates the local models with a strict temporal holdout.

Methods

The final dataset is `data/final/f1_top10_model_dataset.csv`. Each row represents one driver in one race. The label is `top10_finish`, equal to 1 when the final classified result is tenth place or better, and 0 otherwise.

Item	Value
Driver-race observations	6,999
Seasons covered	2010-2026
Final variables	201
Numeric/bool variables	179
Categorical/text variables	22
Missing values after preprocessing	0
Top-10 rows	3,330
Non-top-10 rows	3,669

The main software packages are:

- pandas and numpy for data manipulation
- scikit-learn for preprocessing, pipelines, model training and metrics

- matplotlib and seaborn for EDA figures
- joblib for model serialization
- Plotly for optional neural-network visualization
- python-docx and ReportLab for generated report exports

The main models compared are:

Model	Role in project	Reason for inclusion
Logistic regression	Baseline classifier	Provides a simple interpretable linear benchmark
Random forest	Tree ensemble classifier	Strong tabular baseline and stable rolling-backtest model
Extra trees	Randomized tree ensemble	Tests whether more randomized tree splits improve robustness
Histogram gradient boosting	Champion classifier	Handles non-linear tabular interactions and achieved the best 2025 top-10 score
MLP classifier	Neural-network extension	Tests whether a dense non-linear neural architecture improves classification

The primary evaluation metrics are accuracy, precision, recall, F1, ROC-AUC and race precision@10. Race precision@10 is important because the practical output is a race-level top-10 list rather than only independent row-level labels.

Dataset Preparation and Exploratory Data Analysis

The assignment requires one reasonably large dataset with categorical and numeric data. Although several public sources were used, they are merged into one final modelling dataset. The raw tables are retained for reproducibility, while the final CSV is the single dataset used for modelling.

The dataset is not perfectly clean in its original form. Race APIs use different identifiers, some historical pit-stop data is unavailable before 2011, FastF1 timing coverage is partial, and weather data is an external historical approximation rather than exact FIA sensor data. The preprocessing pipeline handles these issues by using availability flags, deterministic fallbacks, imputation and careful feature merging.

Preparation step	Purpose
Merge raw tables	Combine race results, qualifying, standings, circuits, weather and optional FastF1 features
Create target	Build `top10_finish` from final classified race position
Remove leakage	Exclude post-race columns such as points, final position, fastest lap and same-race race-control counts during training
Use pre-race experience fields	Replace dataset-wide future summaries with race-age and experience counters known before each race
Impute numeric values	Fill missing numeric features with median values
Impute categorical values	Fill missing categorical features with the most frequent category
One-hot encode categories	Convert driver, constructor, circuit and weather categories into model-ready features
Scale numeric values	Standardize numeric variables for logistic regression and neural-network models

Key EDA observations:

- The target is reasonably balanced for a binary classification task: about 48% top-10 rows and 52% non-top-10 rows.
- Rows per season vary because the number of races and drivers changes over time.
- Qualifying/grid position is strongly related to finishing position, but it is not enough alone because reliability, circuit, weather and team performance can change the outcome.
- Categorical fields such as driver, constructor, circuit and weather condition require encoding before model training.

Every figure included in the generated report is discussed in the relevant body section. The assignment-facing PNG figures summarize the full pipeline, holdout algorithm results, validation metric table and expanding-window stability.

Code Organization and Reproducibility

The codebase is organized so that data import, feature engineering, model definition, training and reporting can be inspected separately.

Path	Purpose
`scripts/generate_raw_data.py`	Imports Jolpica race, qualifying, driver, constructor, circuit and pit-stop data
`scripts/rebuild_derived_raw_tables.py`	Rebuilds derived raw features from local event files without refetching
`scripts/generate_historical_weather.py`	Adds Open-Meteo race-day weather features
`scripts/fetch_upcoming_weather_forecast.py`	Creates pre-race weather forecast/fallback snapshots for upcoming races
`scripts/generate_fastf1_features.py`	Adds optional FastF1 timing, weather, tyre and lap-derived features
`scripts/generate_fastf1_race_control.py`	Imports race-control messages and builds disruption indicators
`scripts/generate_final_dataset.py`	Merges all raw and derived tables into the final modelling dataset
`scripts/algorithms/classification.py`	Contains top-10 classifier definitions and hyperparameters
`scripts/algorithms/regression.py`	Contains finish-position ranking model definitions
`scripts/algorithms/neural_network.py`	Contains dedicated MLP neural-network configurations
`scripts/train_model.py`	Trains the selected top-10 classifier and saves metrics/figures
`scripts/evaluate_models.py`	Compares all classifiers and runs rolling backtests
`scripts/tune_neural_network.py`	Tunes the dedicated MLP classifier
`scripts/train_position_model.py`	Trains the finish-position ranking models
`scripts/build_report_docx.py`	Generates the Word report
`scripts/build_report_pdf.py`	Generates the PDF report
`scripts/validate_project.py`	Checks that required files, outputs and metrics exist

The main reproducibility commands are:

```
python scripts/generate_final_dataset.py
python scripts/rebuild_derived_raw_tables.py
python scripts/fetch_upcoming_weather_forecast.py --season 2026
```

```
--count 4
python scripts/evaluate_models.py
python scripts/train_model.py --model random_forest
python scripts/tune_neural_network.py --force
python scripts/train_position_model.py
python scripts/make_charts.py
python scripts/build_report_docx.py
python scripts/build_report_pdf.py
python scripts/build_submission.py
python scripts/validate_project.py
```

Code comments are kept concise and focused. The main documentation is provided through `README.md`, `docs/ALGORITHMS.md`, `docs/ASSIGNMENT_ALIGNMENT.md`, the notebook and this report, so the submitted code remains readable without turning the report into a large code listing.

Model Implementation

For the main classification task, the selected champion model is random forest. This model was selected because it achieved the best 2025 holdout race precision@10 after the leakage-safe feature cleanup, while still handling non-linear interactions in tabular data well. Extra trees and histogram gradient boosting remain important comparison models because they are close on rolling validation and latest-season performance.

Model	Important parameters	Notes
Logistic regression	`max_iter=1500`, `class_weight=balanced`	Baseline model with scaled numeric features
Random forest	`n_estimators=500`, `min_samples_leaf=3`	Main champion classifier
Extra trees	`n_estimators=500`, `min_samples_leaf=3`	More randomized tree ensemble
Histogram gradient boosting	`max_iter=300`, `learning_rate=0.05`, `max_leaf_nodes=31`, `l2_regularization=0.05`	Strong boosted-tree comparison
MLP classifier	hidden layers tested from `(64, 32)` to `(128, 64, 32)`	Additional neural-network experiment

As an additional experiment beyond the two required predictive models, a Multi-Layer Perceptron neural network was implemented using scikit-learn's `MLPClassifier`. The purpose was to test whether a non-linear neural architecture could capture more complex relationships in the dataset compared with traditional machine learning models. Several neural configurations were tested by varying the number of hidden layers, number of neurons, activation function, learning rate and L2 regularization parameter. Early stopping was also used to reduce overfitting by monitoring validation performance during training.

The MLP model used dense numerical input after preprocessing. Numerical variables were scaled, categorical variables were encoded, and the final transformed feature matrix was passed to the neural network. Since neural networks are sensitive to feature scale, standardization was applied to numerical features. The dedicated tuned MLP experiment uses hidden layers `(80, 40)`, tanh activation, `alpha=0.002`, `learning_rate_init=0.0007`, `batch_size=64` and early stopping. The baseline MLP included in the main algorithm comparison uses the shared classifier configuration from `scripts/algorithms/classification.py`. These neural-network models are useful non-linear baselines, but they are not presented as the main assignment model because the tree-based methods remain stronger or more stable on the tabular prediction task.

Model Validation

A temporal validation strategy is used instead of a random split. This is more realistic because a model should learn from past seasons and predict future races. The main holdout test season is 2025. An expanding-window rolling backtest is also used to compare stability across seasons.

Current holdout results on the 2025 season:

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Race precision@10
Random forest	0.768	0.753	0.800	0.776	0.842	0.775
Histogram gradient boosting	0.768	0.768	0.771	0.769	0.835	0.771
Extra trees	0.752	0.739	0.779	0.759	0.836	0.771
Neural network MLP	0.770	0.810	0.708	0.756	0.828	0.758
Logistic regression	0.739	0.836	0.596	0.696	0.820	0.750

Training-time measurements on the current machine show that a single holdout run is lightweight. The heavier steps are the rolling backtest and full neural-network tuning because they train several models repeatedly.

Algorithm	Single holdout run time
Logistic regression	0.35 s
Random forest	1.66 s
Extra trees	2.05 s
Neural network MLP	2.46 s
Histogram gradient boosting	9.00 s

Rolling backtest averages show that extra trees is the strongest model by race precision@10, with an average of 0.770, while random forest is very close at 0.769 and remains the latest-season holdout champion.

Race-Level Prediction Renders

To make the model outputs more concrete, an additional render script generates race-level prediction artifacts for the 2025 holdout season. These outputs do not replace the statistical validation, but they make the model behaviour easier to inspect race by race.

Output	Purpose
`outputs/predictions/race_model_renders/race_model_rankings_2025.csv`	Full driver-level predicted ranking for each race and each model
`outputs/predictions/race_model_renders/race_model_summary_2025.csv`	Race-level summary with top-10 hits, actual points captured, podium hits and heuristic explanations
`outputs/figures/predictions/model_precision_by_race.png`	Line chart showing correct top-10 hits per model and race
`outputs/figures/predictions/model_hit_heatmap.png`	Heatmap comparing top-10 hits by race and model
`outputs/figures/predictions/model_points_captured.png`	Average actual points captured by each model's predicted top 10
`outputs/figures/predictions/race_overviews/*.png`	One-page race overviews with real top 10, model scoreboard, virtual podiums and consensus picks
`outputs/figures/predictions/race_cards/*.png`	Visual race cards with actual podium, virtual podiums and predicted top 10 lists

The race overview and race card outputs show each model's virtual podium and predicted top 10 for a completed race, alongside the real top-10 result. Green rows or chips indicate drivers who actually finished in the real top 10, while red rows or chips indicate predicted top-10 drivers who missed the real top 10. The points shown are actual race points earned after the race. Driver headshots are retrieved through OpenF1's `headshot_url` metadata when available; if a photo cannot be retrieved, the render falls back to a clean initials-based placeholder.

These visual outputs are useful for explaining why one model performed better on a specific race. For example, a model may win a race-level comparison because it captured more midfield drivers who finished in the points, avoided overrating front-grid drivers who later dropped out, or produced a virtual podium closer to the real podium.

Analysis and Recommendations

The results are broadly in line with expectations. Tree-based models perform strongly because the dataset is tabular, mixed-type and feature-engineered. Logistic regression is useful as a simple baseline but is less able to capture non-linear interactions such as driver-team-circuit effects. The neural-network classifier is competitive but does not clearly beat tree ensembles on this dataset. This supports the decision to present the neural network as an extension rather than the primary assignment result.

Adding the 2010 season increased the dataset from 6,543 to 6,999 rows. A later feature refresh then added richer pit-stop aggregates, more FastF1 lap-summary coverage where the API allowed it, and a separate upcoming-race weather snapshot table. After replacing dataset-wide future summaries with pre-race age and experience features, the best leakage-safe 2025 race precision@10 is 0.775. This is slightly lower than the earlier draft result, but it is a cleaner and more defensible estimate because the model no longer sees future-known experience summaries.

The separate finish-position ranking model is useful for producing ordered race predictions. Its best current version is a histogram gradient boosting regressor with race precision@10 of 0.775 and mean race Spearman correlation of 0.653. This is helpful for interpretation, but it should remain secondary because the assignment asks mainly for predictive model comparison.

Recommended final model:

- main assignment model: random forest classifier
- supporting baselines: extra trees and histogram gradient boosting classifiers
- optional extension: tuned MLP classifier and finish-position regressor

Generated report deliverables:

- `report/Report.md`: editable report source
- `report/Report.docx`: Word report version
- `report/Report.pdf`: PDF report version generated from the same source
- `outputs/figures/*.png`: labelled EDA, algorithm and validation figures used as report evidence
- `submission/IML_Assignment_GroupX.zip`: compressed submission archive

Recommended future improvements:

- add calibration plots for probability quality
- add permutation importance for model interpretability
- fill remaining FastF1 gaps after rate limits reset
- rerun upcoming weather snapshots near race week so they use real forecast data
- test whether older seasons should be down-weighted because modern F1 regulations differ
- improve the report with more domain-specific literature if more time is available

Conclusion

This project satisfies the core assignment requirement by building a machine learning solution on one reasonably large, mixed-type and non-trivial dataset. The work includes data collection, cleaning, feature engineering, EDA, preprocessing, multiple predictive models, tuning, validation and comparison. The strongest model for the main top-10 classification task is random forest, with 2025 holdout race precision@10 of 0.775. The project also shows that neural networks are interesting but not automatically superior for this tabular problem. The main weakness is that some racing signals, especially full telemetry, complete FastF1 historical coverage and exact race-week weather forecasts, are incomplete in public data. Overall, the project is a solid V0 for the assignment, with neural-network and 3D visualization work best positioned as optional extensions.

Acknowledgements

This project uses publicly available motorsport and weather data from Jolpica, FastF1 and Open-Meteo. The implementation relies on open-source Python libraries including pandas, numpy, scikit-learn, matplotlib, seaborn, Plotly, joblib, python-docx and ReportLab. The assignment structure follows the CX016-2.5-3-IML group assignment brief provided in the local `docs/` folder.

References

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Appendix: Neural Network and Ranking Extensions

The neural-network work is included as an extension rather than the main assignment claim. The binary MLP classifier is implemented in `scripts/algorithms/neural_network.py` and trained through `scripts/tune_neural_network.py`. The 3D neural visualization is generated by `scripts/visualize_neural_network_3d.py` and exported to `outputs/figures/neural_network_embedding_3d.html`. A static report-friendly image is also exported to `outputs/figures/neural_network_embedding_3d.png`.

The finish-position ranking extension is implemented in `scripts/train_position_model.py` and `scripts/algorithms/regression.py`.

It predicts an ordered finishing rank for each driver and is used in upcoming-race prediction exports. This extension is useful for practical race prediction but should be treated as secondary to the assignment's main supervised classification task.

Figures

Figure 1. End-to-end project pipeline from raw data sources to validated assignment outputs.

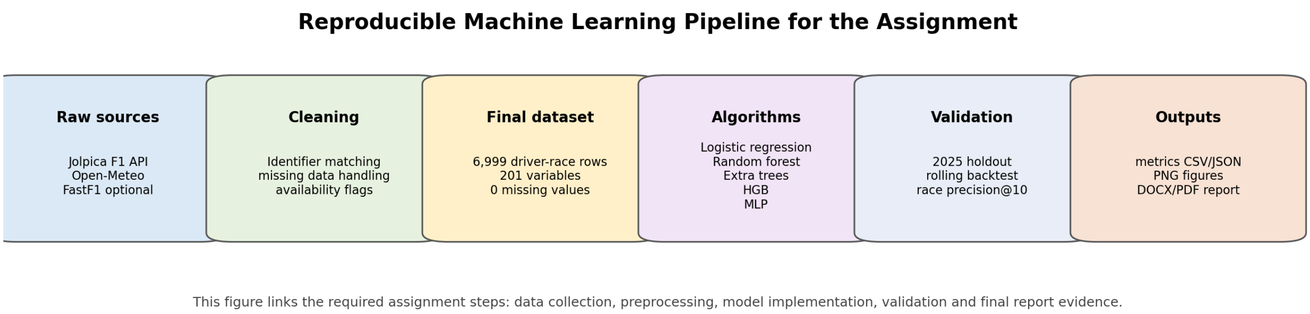

Figure 2. Target distribution for the binary top-10 classification task.

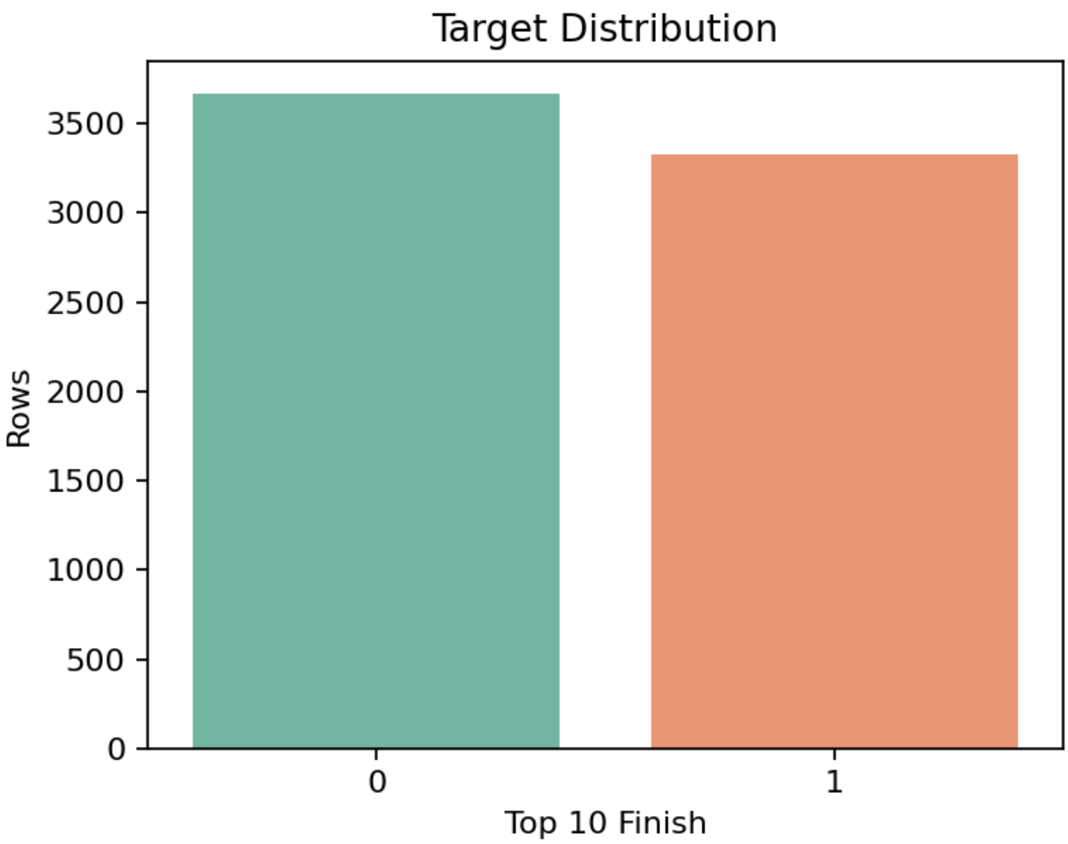

Figure 3. Number of driver-race rows available per season.

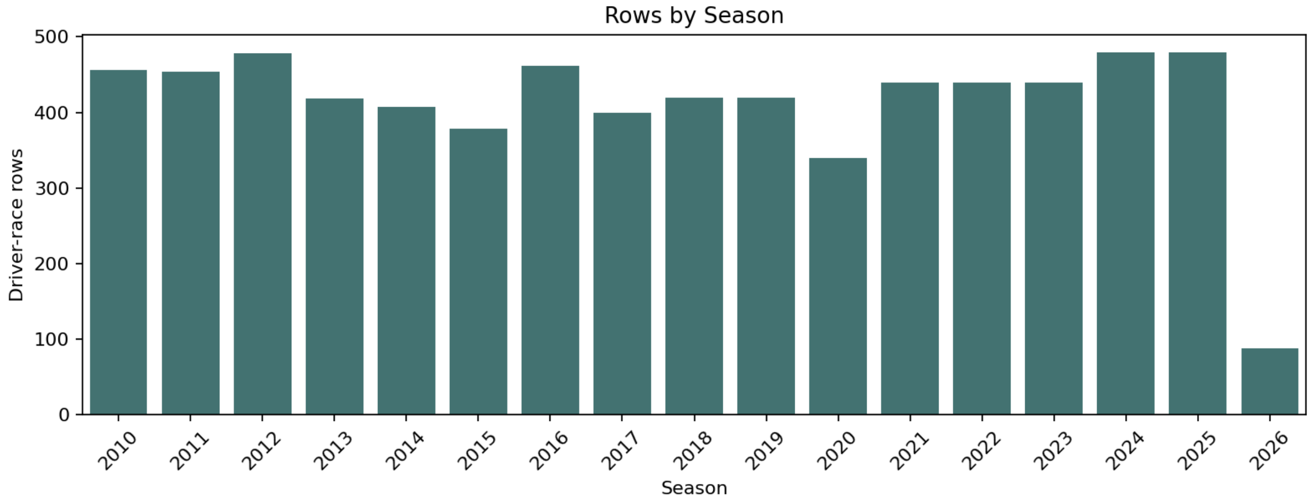

Figure 4. Top-10 target rate by season, used to check class stability over time.

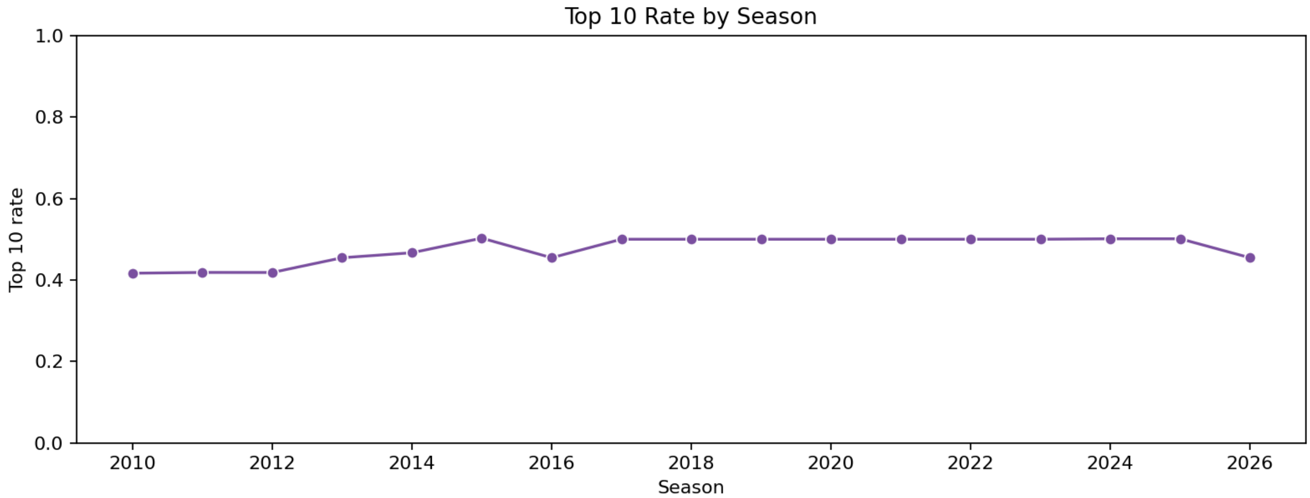

Figure 5. Relationship between starting grid and final position.

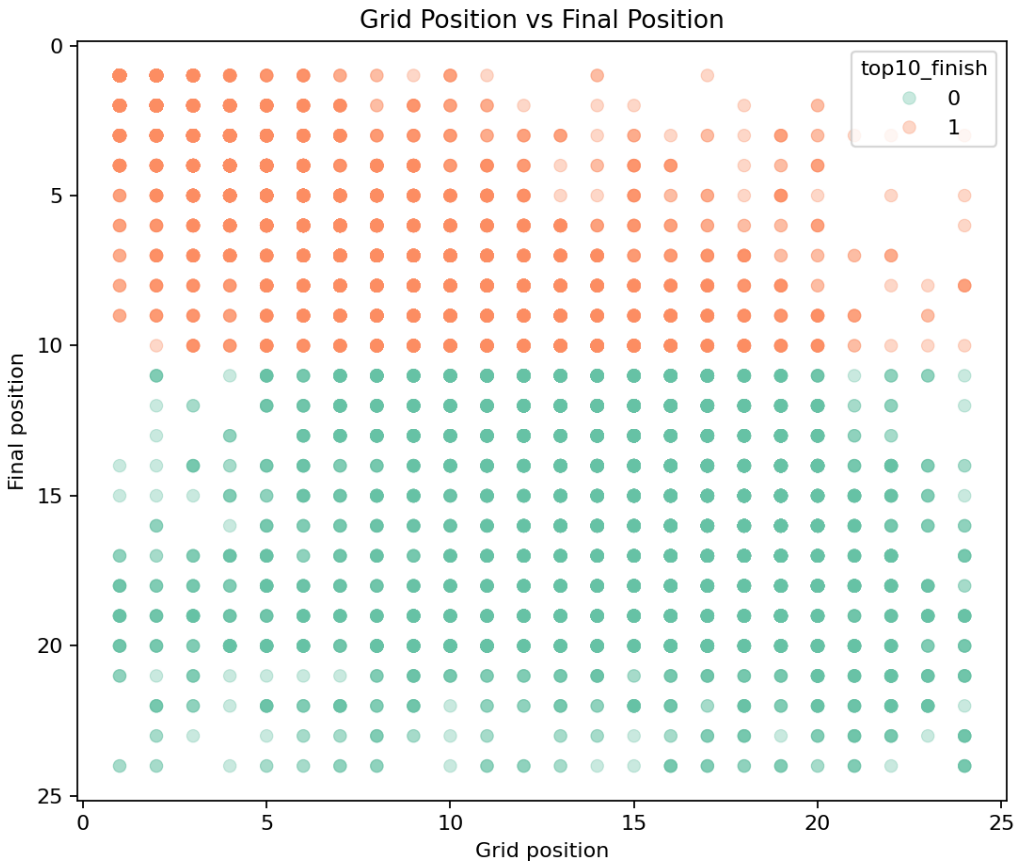

Figure 6. Holdout comparison of the main classification algorithms on the 2025 season.

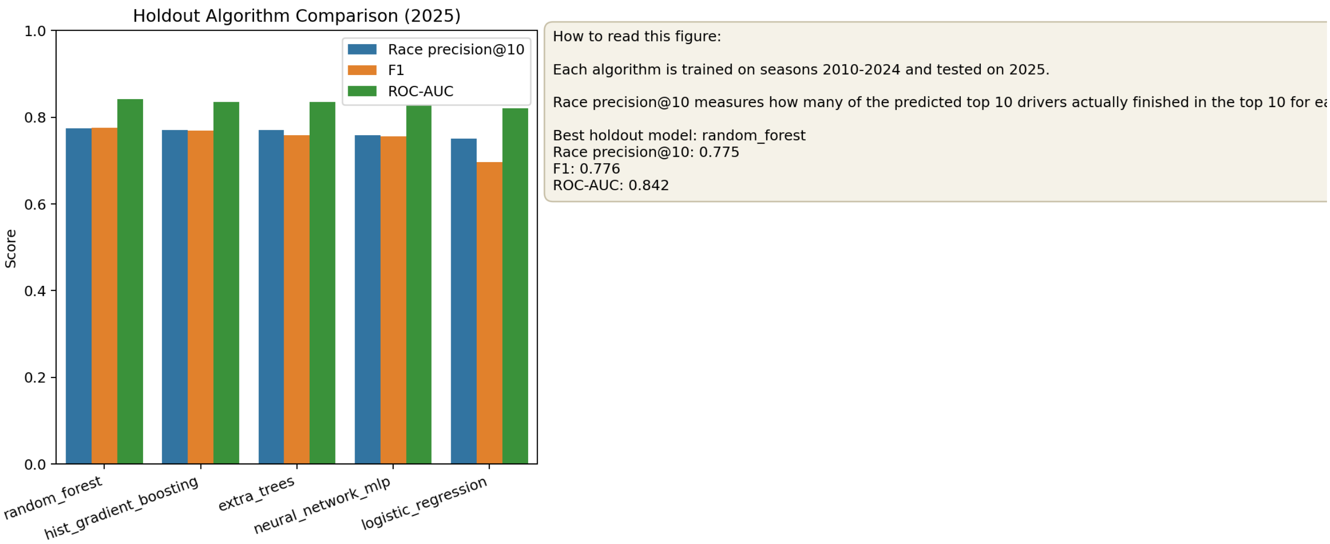

Figure 7. Validation metric table for the compared classification algorithms.

Model Validation Metrics on 2025 Holdout

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Race P@10
random_forest	0.768	0.753	0.800	0.776	0.842	0.775
hist_gradient_boosting	0.768	0.768	0.771	0.769	0.835	0.771
extra_trees	0.752	0.739	0.779	0.759	0.836	0.771
neural_network_mlp	0.770	0.810	0.708	0.756	0.828	0.758
logistic_regression	0.739	0.836	0.596	0.696	0.820	0.750

The table is sorted by race precision@10, the metric most aligned with predicting a race-level top 10.

Figure 8. Compact model comparison across F1, ROC-AUC and race precision@10.

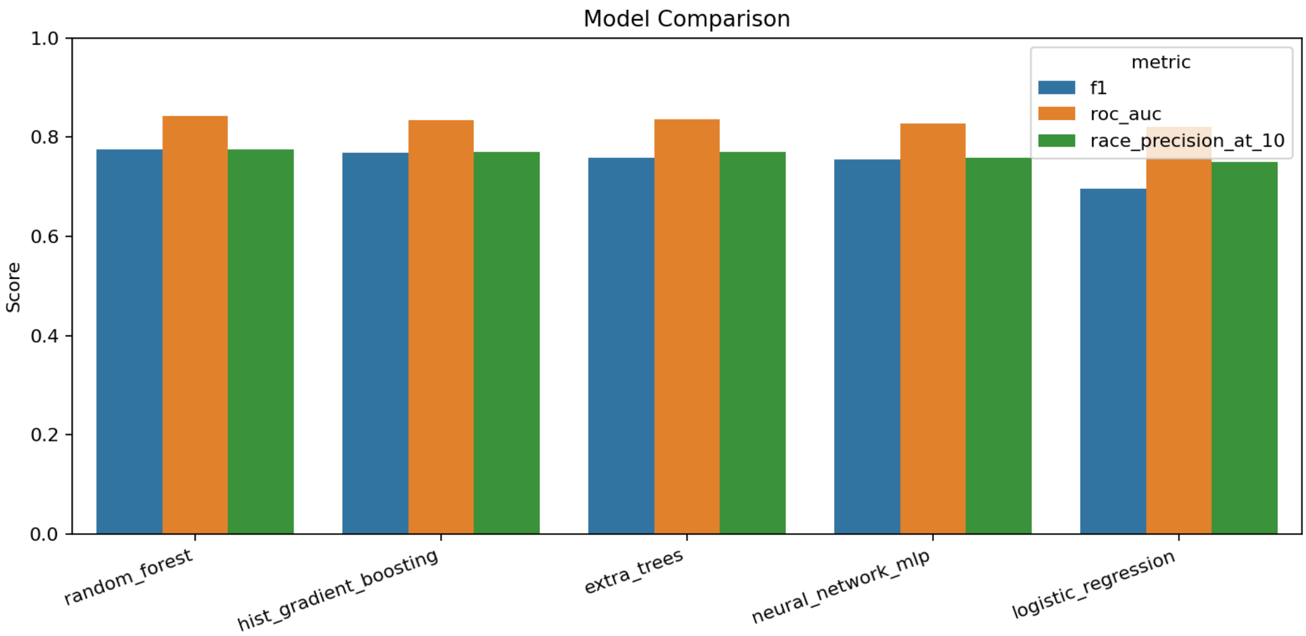

Figure 9. Expanding-window validation showing model stability across seasons.

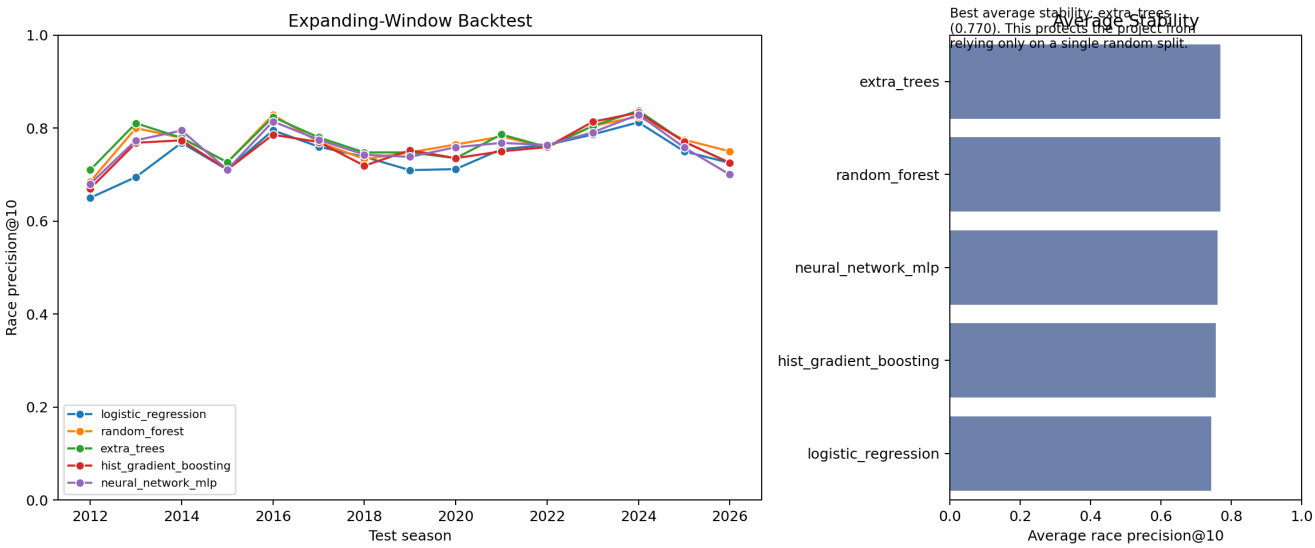

Figure 10. Rolling backtest race precision@10 by model and test season.

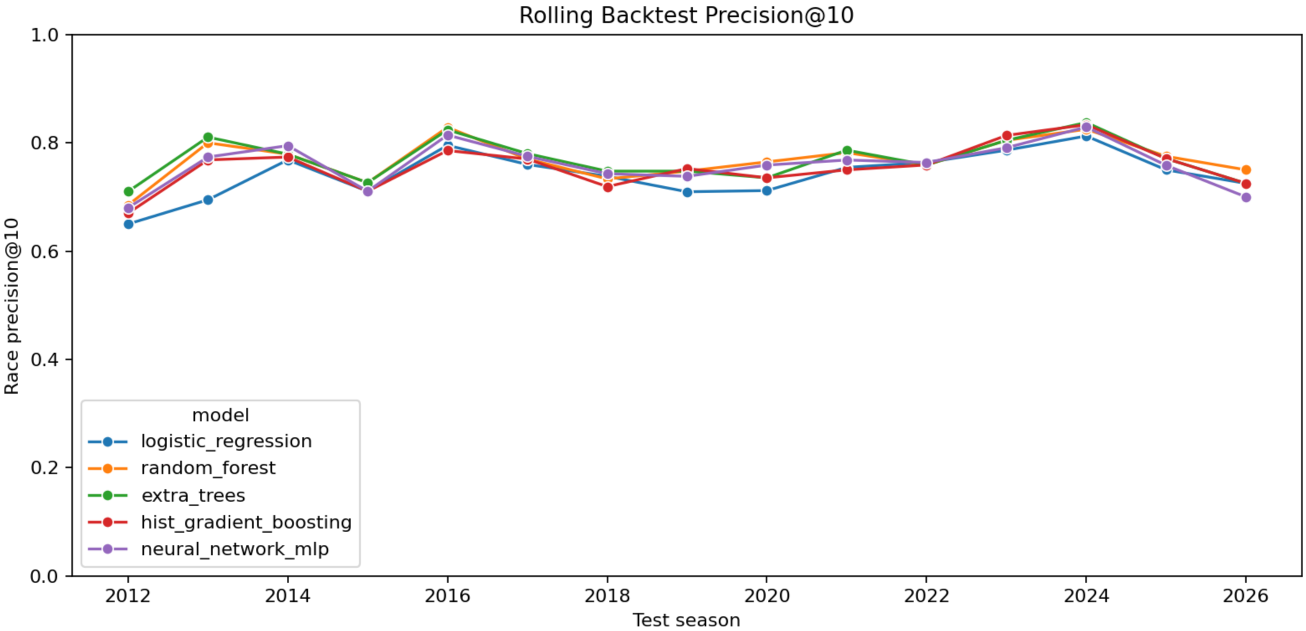

Figure 11. Most influential features for the selected top-10 classifier.

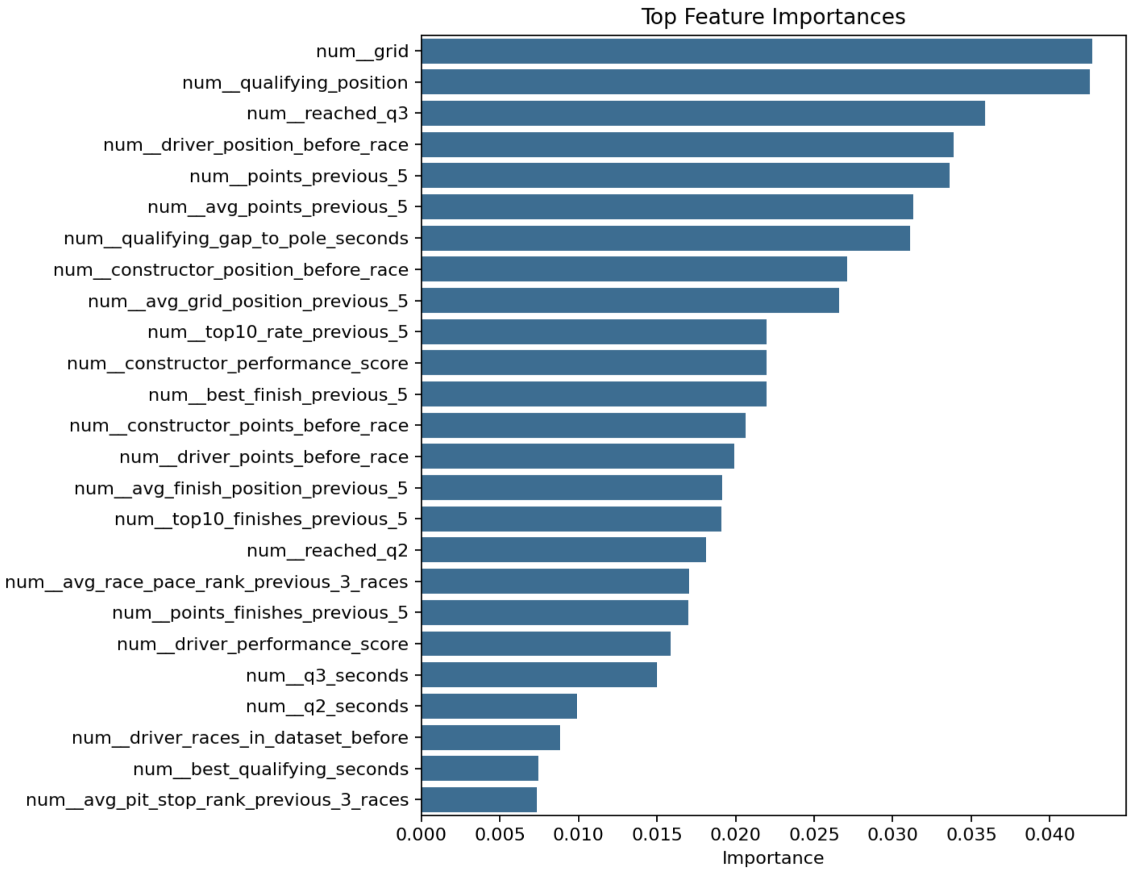

Figure 12. Confusion matrix for the selected holdout classifier.

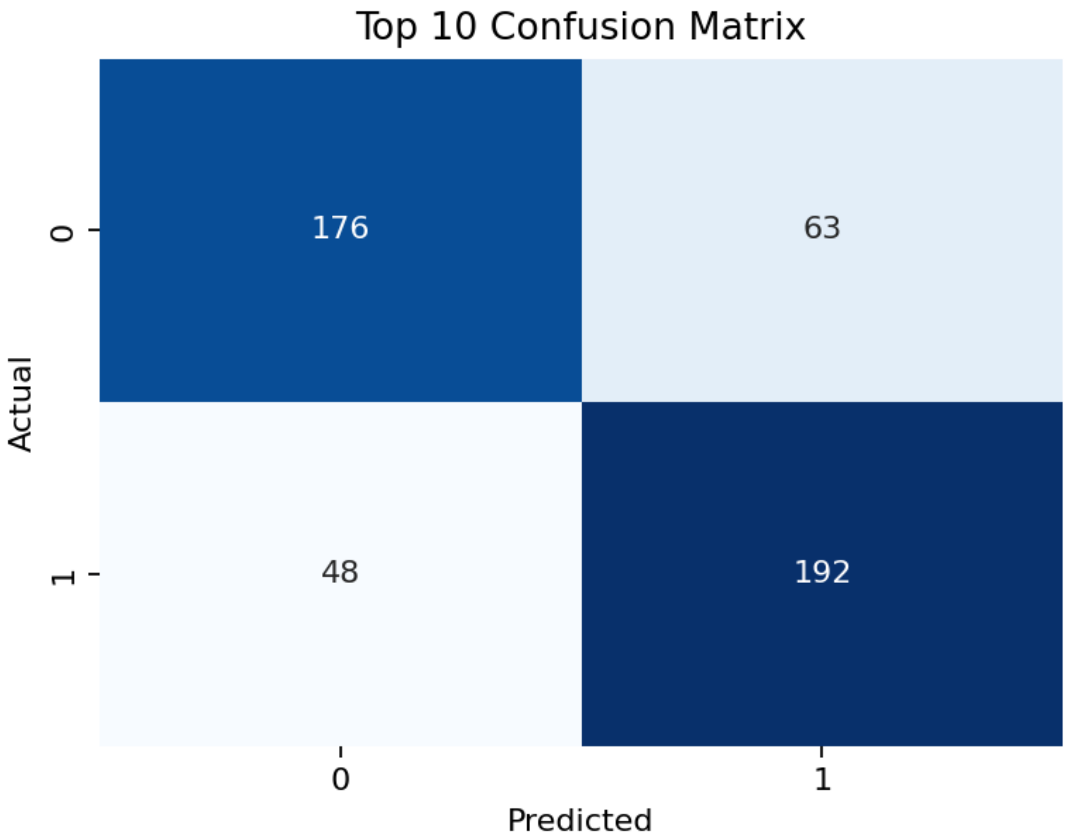

Figure 13. Neural-network classifier tuning results, included as an extension.

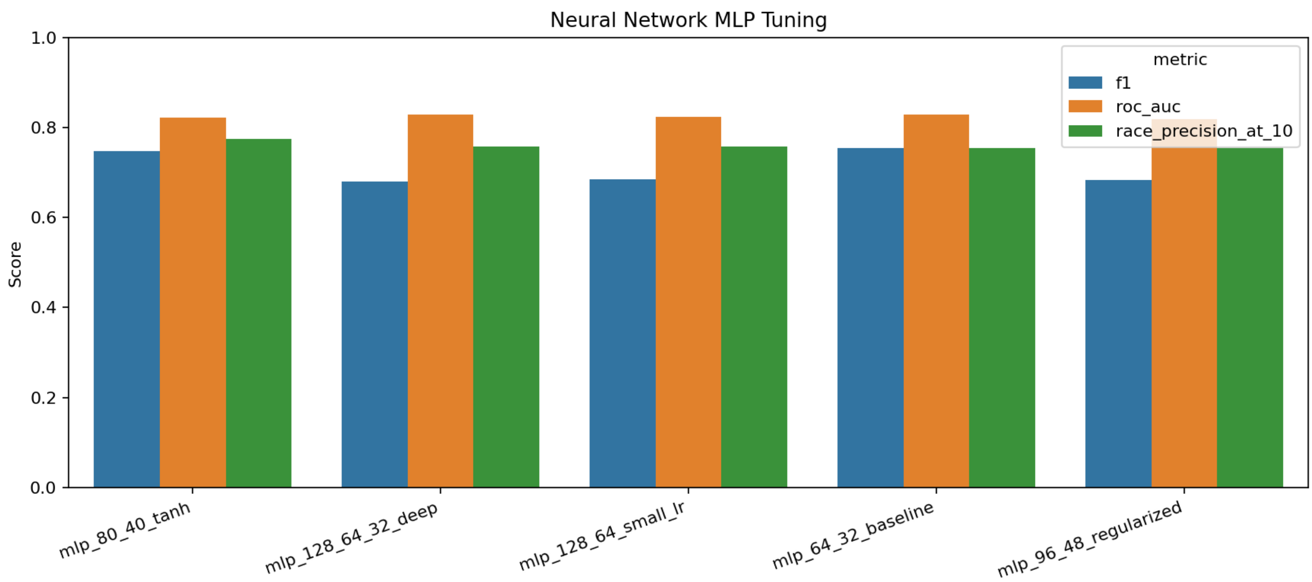

Figure 14. Static view of the interactive 3D neural-network hidden-space embedding.

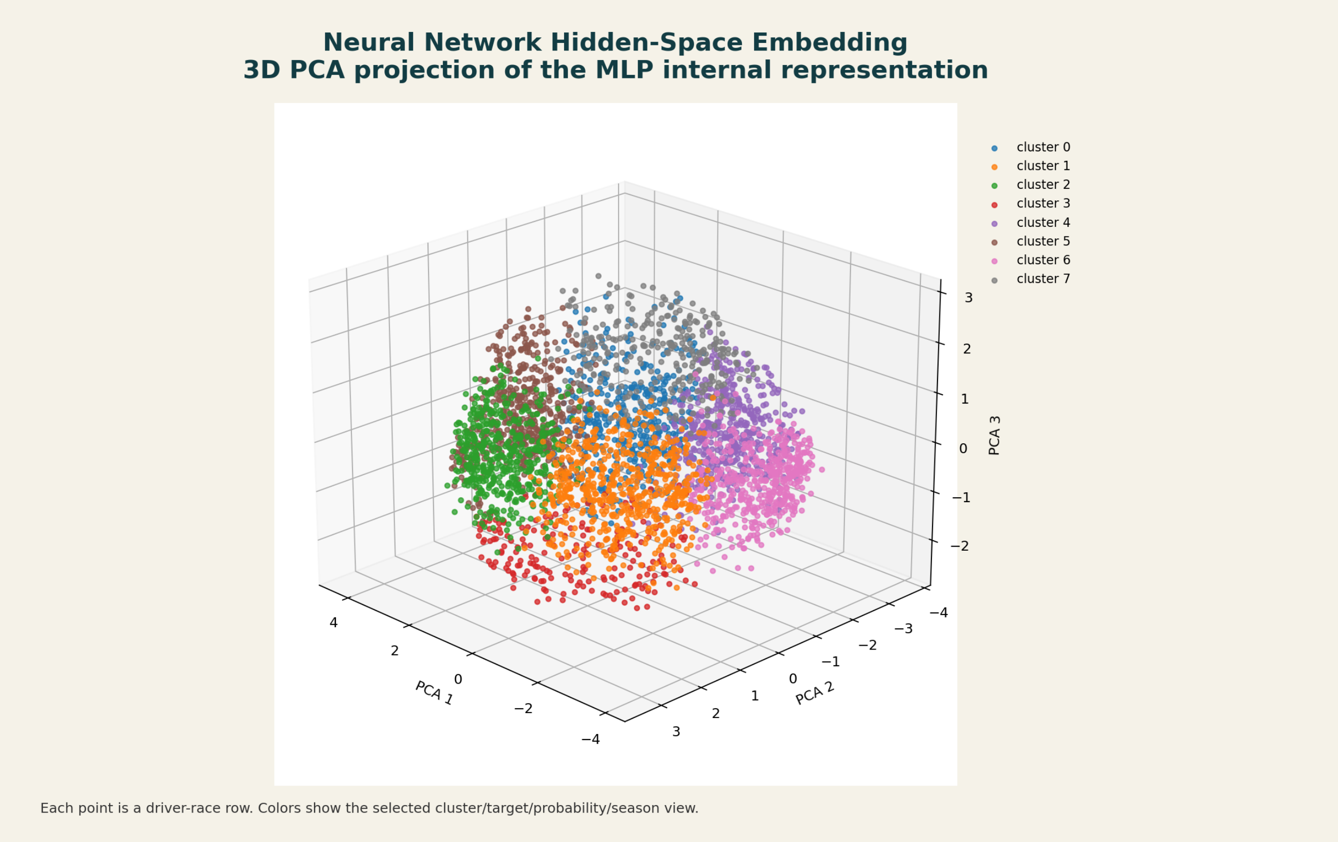

Figure 15. Finish-position ranking model comparison, included as an extension.

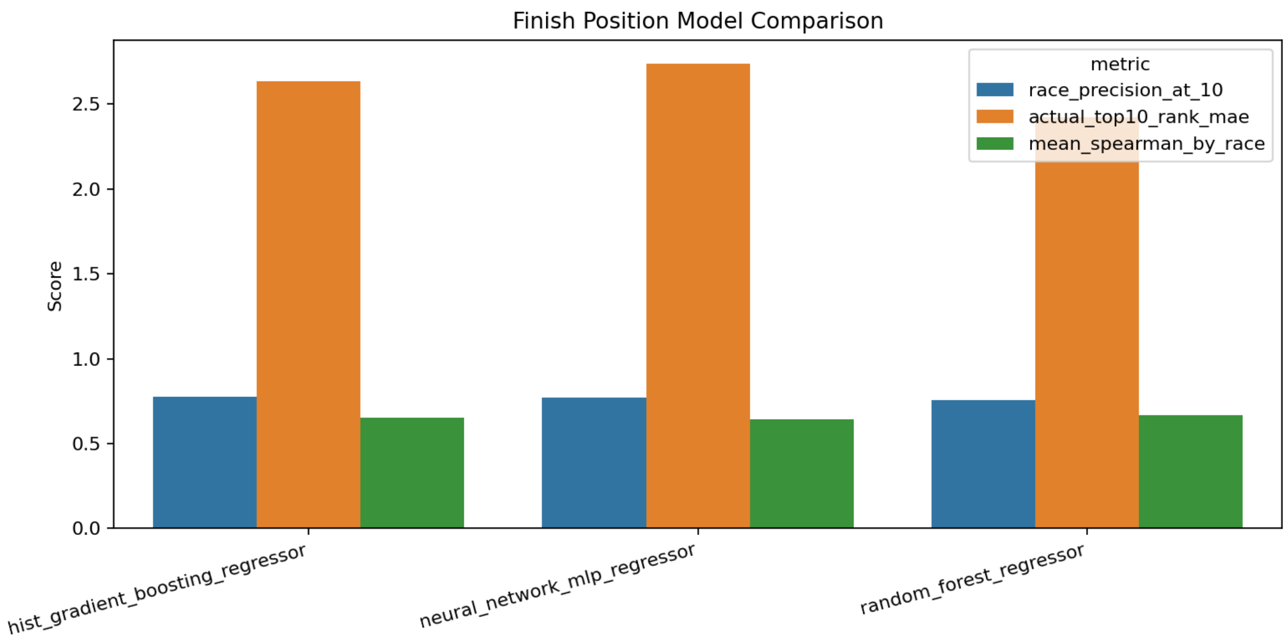

Figure 16. Correct predicted top-10 drivers per race and model on the 2025 holdout season.

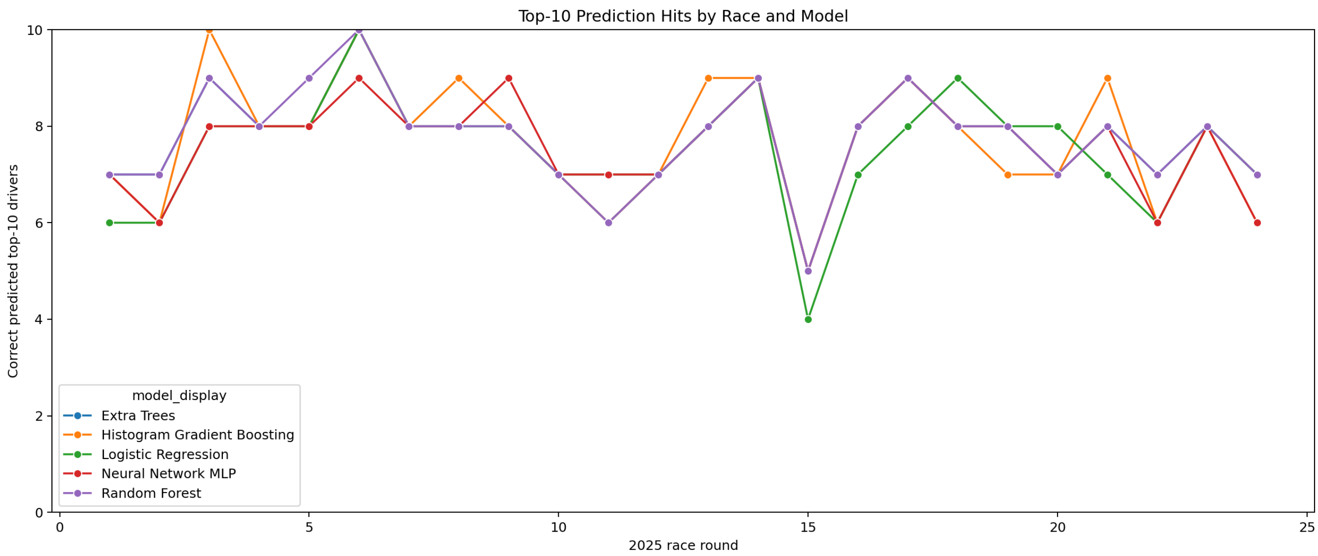

Figure 17. Race-level heatmap of top-10 hits by model.

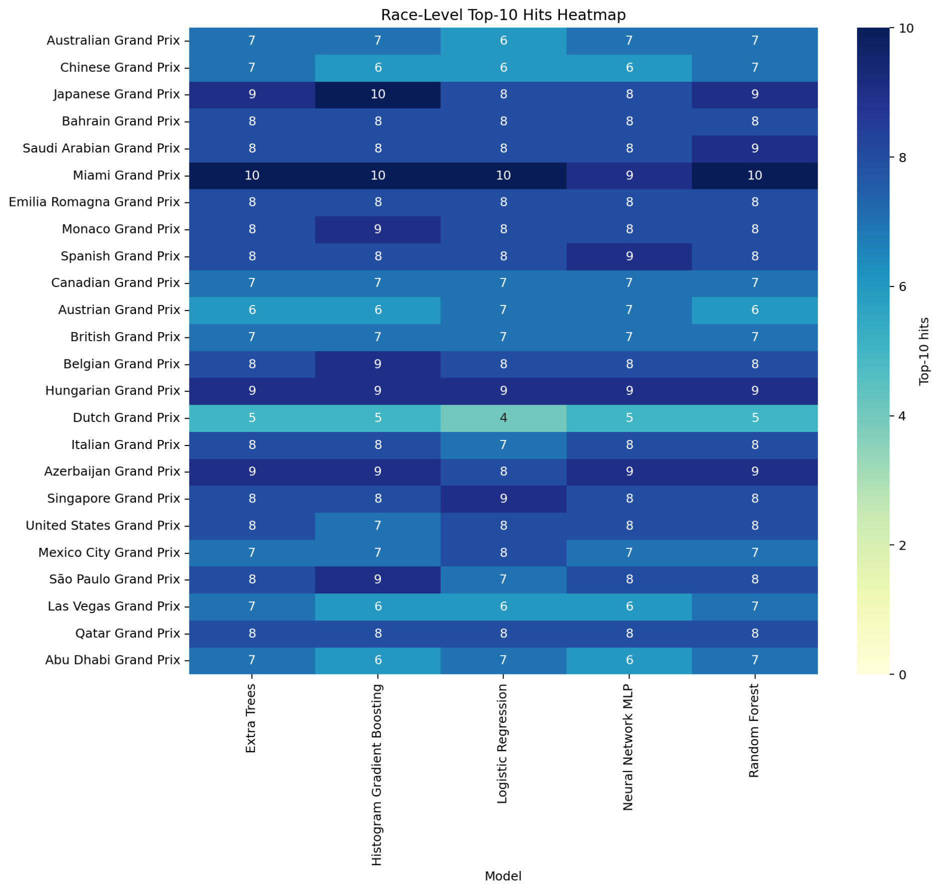

Figure 18. Average actual points captured by each model's predicted top 10.

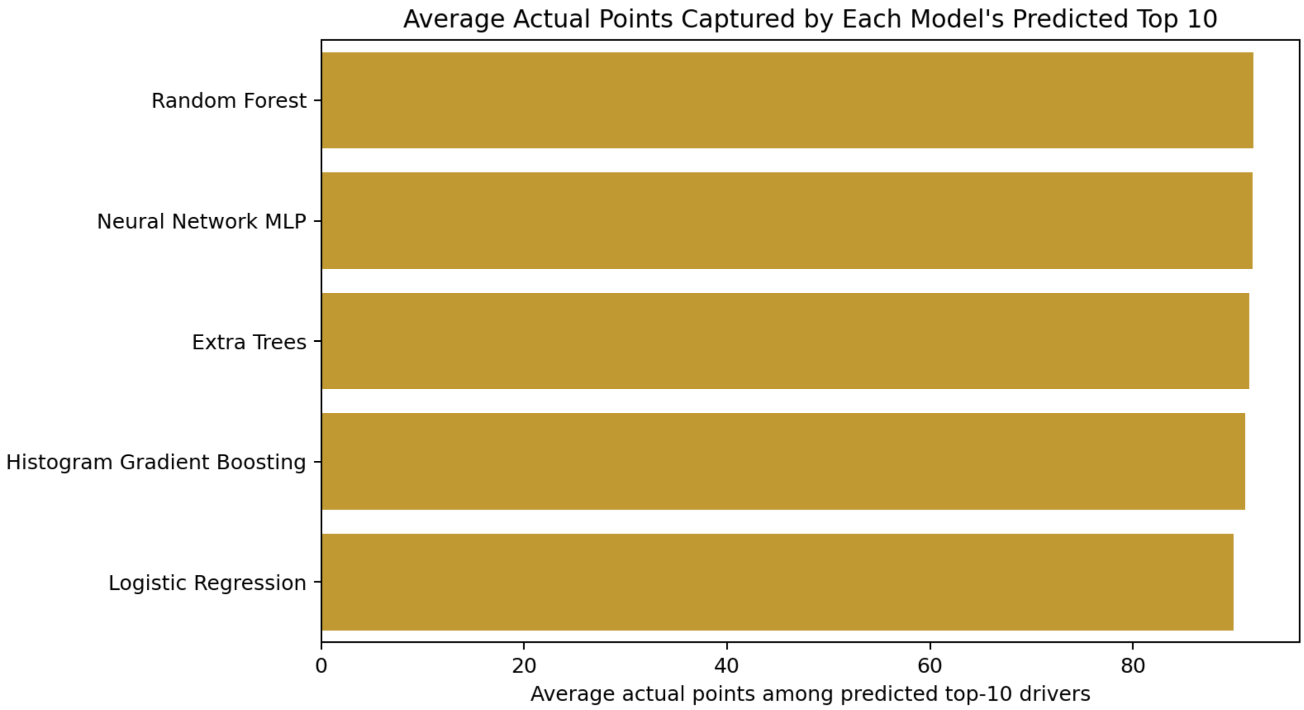

Figure 19. Example race overview with real result, model scoreboard, consensus picks and predicted top 10 chips.

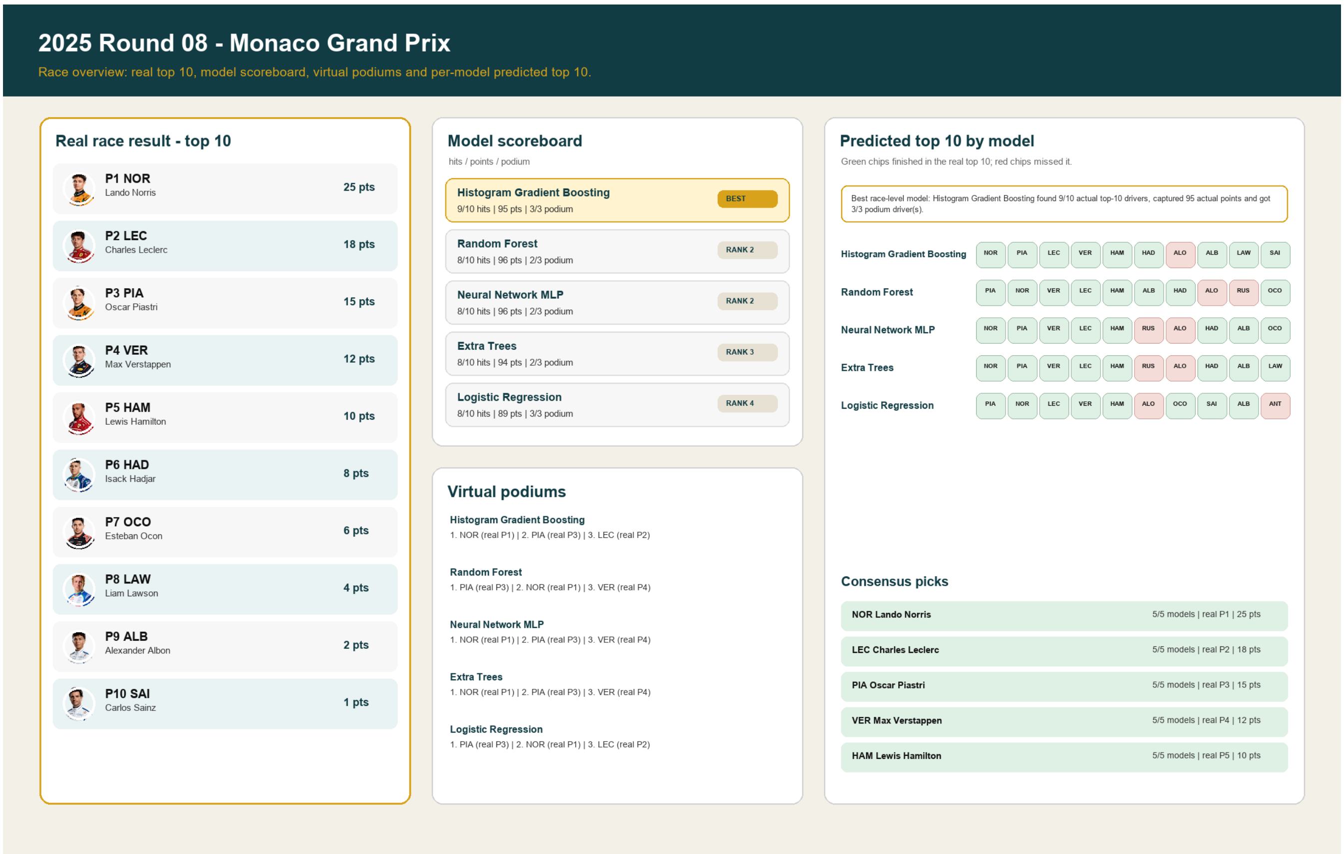

Figure 20. Detailed race card showing actual podium, virtual podiums and predicted top 10 lists.

2025 Round 08 - Monaco Grand Prix

Models are sorted by this race's score. Green rows finished top 10; red rows missed it; points are actual race points.

Actual podium

P1 NOR
Lando Norris
25 pts

P2 LEC
Charles Leclerc
18 pts

P3 PIA
Oscar Piastri
15 pts

Histogram Gradient Boosting

BEST

9/10 hits | 95 pts | 3/3 podium

#1 NOR
Actual P1 | 25 pts | 98%

#2 PIA
Actual P3 | 15 pts | 97%

#3 LEC
Actual P2 | 18 pts | 95%

Predicted top 10		prob	real	pts
01	NOR Lando Norris	98%	P1	25
02	PIA Oscar Piastri	97%	P3	15
03	LEC Charles Leclerc	95%	P2	18
04	VER Max Verstappen	95%	P4	12
05	HAM Lewis Hamilton	91%	P5	10
06	HAD Isack Hadjar	87%	P6	8
07	ALO Fernando Alonso	79%	P19	0
08	ALB Alexander Albon	77%	P9	2
09	LAW Liam Lawson	67%	P8	4
10	SAI Carlos Sainz	50%	P10	1

Random Forest

RANK 2

8/10 hits | 96 pts | 2/3 podium

#1 PIA
Actual P3 | 15 pts | 91%

#2 NOR
Actual P1 | 25 pts | 90%

#3 VER
Actual P4 | 12 pts | 90%

Predicted top 10		prob	real	pts
01	PIA Oscar Piastri	91%	P3	15
02	NOR Lando Norris	90%	P1	25
03	VER Max Verstappen	90%	P4	12
04	LEC Charles Leclerc	89%	P2	18
05	HAM Lewis Hamilton	85%	P5	10
06	ALB Alexander Albon	69%	P9	2
07	HAD Isack Hadjar	63%	P6	8
08	ALO Fernando Alonso	61%	P19	0
09	RUS George Russell	60%	P11	0
10	OCO Esteban Ocon	53%	P7	6

Neural Network MLP

RANK 2

8/10 hits | 96 pts | 2/3 podium

#1 NOR
Actual P1 | 25 pts | 90%

#2 PIA
Actual P3 | 15 pts | 86%

#3 VER
Actual P4 | 12 pts | 82%

Predicted top 10		prob	real	pts
01	NOR Lando Norris	90%	P1	25
02	PIA Oscar Piastri	86%	P3	15
03	VER Max Verstappen	82%	P4	12
04	LEC Charles Leclerc	80%	P2	18
05	HAM Lewis Hamilton	79%	P5	10
06	RUS George Russell	59%	P11	0
07	ALO Fernando Alonso	57%	P19	0
08	HAD Isack Hadjar	54%	P6	8
09	ALB Alexander Albon	54%	P9	2
10	OCO Esteban Ocon	50%	P7	6

Extra Trees

RANK 3

8/10 hits | 94 pts | 2/3 podium

#1 NOR
Actual P1 | 25 pts | 94%

#2 PIA
Actual P3 | 15 pts | 92%

#3 VER
Actual P4 | 12 pts | 91%

Predicted top 10		prob	real	pts
01	NOR Lando Norris	94%	P1	25
02	PIA Oscar Piastri	92%	P3	15
03	VER Max Verstappen	91%	P4	12
04	LEC Charles Leclerc	88%	P2	18
05	HAM Lewis Hamilton	84%	P5	10
06	RUS George Russell	72%	P11	0
07	ALO Fernando Alonso	69%	P19	0
08	HAD Isack Hadjar	66%	P6	8
09	ALB Alexander Albon	63%	P9	2
10	LAW Liam Lawson	59%	P8	4

Logistic Regression

RANK 4

8/10 hits | 89 pts | 3/3 podium

#1 PIA
Actual P3 | 15 pts | 91%

#2 NOR
Actual P1 | 25 pts | 90%

#3 LEC
Actual P2 | 18 pts | 82%

Predicted top 10		prob	real	pts
01	PIA Oscar Piastri	91%	P3	15
02	NOR Lando Norris	90%	P1	25
03	LEC Charles Leclerc	82%	P2	18
04	VER Max Verstappen	79%	P4	12
05	HAM Lewis Hamilton	71%	P5	10
06	ALO Fernando Alonso	58%	P19	0
07	OCO Esteban Ocon	52%	P7	6
08	SAI Carlos Sainz	45%	P10	1
09	ALB Alexander Albon	45%	P9	2
10	ANT Andrea Kimi Antonelli	43%	P18	0